

Study Questions: Learning

1. Distinguish parameter learning from structure learning. What neural mechanisms implement each?
2. What is long-term potentiation (LTP)? How does Hebb's rule ("fire together, wire together") relate to Active Inference?
3. Explain spike-timing-dependent plasticity (STDP). How does the temporal relationship between pre- and post-synaptic firing encode prediction error?
4. How does Bayesian Model Reduction (BMR) differ from simple parameter updating? Why is it computationally efficient?
5. How does the synaptic homeostasis hypothesis (Tononi & Cirelli) relate to structure learning during sleep?
6. What happens during NREM sleep that might implement Bayesian Model Reduction? What neural evidence supports this?
7. What is a critical period, and how does Active Inference explain why plasticity is heightened and then reduced?
8. How does the concept of precision change across development? Compare infant and adult precision profiles.
9. How does expertise develop in terms of the generative model? What changes in precision, hierarchy, and habitual policies?
10. How does Active Inference account for the phenomenon of "overlearning" — performing a skill so automatically that conscious attention degrades performance?
11. What goes wrong in Alzheimer's disease from a synaptic plasticity perspective? How does this manifest as failures of inference?
12. How does PTSD represent maladaptive learning? What high-precision priors are created by traumatic experience?
13. How does exposure therapy work in Active Inference terms? What precision dynamics are being modified?
14. Compare Active Inference's account of learning with classical conditioning (Pavlov). How does prediction error drive conditioning?
15. What is the role of the hippocampus in learning within the Active Inference framework?

16. How does neurogenesis (new neuron growth) in the hippocampus relate to structure learning?
17. What is the "stability-plasticity dilemma"? How does the brain balance maintaining stable models with remaining open to new information?
18. How does reinforcement learning (Rescorla-Wagner model) relate to Active Inference's account of prediction error-driven learning?
19. Can unlearning occur deliberately? How does Active Inference model the extinction of conditioned responses?
20. Design a sleep study that could test whether a night of sleep improves model simplicity (as BMR predicts) rather than simply strengthening memories.